Project 2: Dataset Description

Student Exam Performance Dataset (Nonlinear Version)

Context

This dataset contains records from a university course tracking student study habits and exam outcomes. Each row represents one student's data for a single exam period.

Important difference from Project 1: This dataset captures a more realistic scenario where the relationship between features and outcomes is **not purely linear**. For example, studying more hours generally helps — but beyond a certain point, additional study hours may lead to diminishing returns or even burnout. Similarly, the combination of study hours and sleep quality may matter more than either feature alone.

Files

- `p2_train.csv` — Training data for model development. This data may contain quality issues that reflect real-world data collection challenges.
- `p2_test.csv` — Held-out test data for final model evaluation. Do not use this file during training or preprocessing parameter fitting.

Column Descriptions

Column Name	Data Type	Description
`student_id`	Integer	Unique identifier assigned to each student record.
`hours_studied`	Float	Self-reported number of hours the student spent studying for the exam. Typical range: 1–12 hours.
`sleep_hours`	Float	Average hours of sleep per night during the exam preparation period. Typical range: 3–10 hours.
`attendance_rate`	Float	Percentage of lectures attended during the course. Reported as a percentage (e.g., 85.5 means 85.5%).
`prev_exam_score`	Float	The student's score on the previous exam in the same course. Scale: 0–100.
`lucky_number`	Integer	A number reported by each student in a pre-exam survey.
`exam_score`	Float	The student's score on the current exam. Scale: 0–100.
`passed`	Integer (0 or 1)	Whether the student passed the exam (1 = passed, 0 = failed). **Classification target for this project.**

Target Variable

This project focuses on **classification**: predicting `passed` (binary label: 0 or 1).

You may use `exam_score` for exploratory analysis, but the primary prediction task is pass/fail classification.

Important Notes

- The training data may contain data quality issues. Exploring and addressing these issues is an expected part of your workflow.
- Not all columns may be useful as input features for prediction. Consider each column carefully before including it in your model.
- **The pass/fail boundary in this dataset is nonlinear.** A simple linear decision boundary (e.g., from Logistic Regression) is expected to perform noticeably worse than nonlinear models. This is by design — the purpose of this project is to explore when and why nonlinear methods outperform linear ones.
- The test set is meant for final evaluation only. Any preprocessing parameters (e.g.,

mean, standard deviation for normalization) should be computed from the training set and then applied to the test set.